



Original article

Mathematical modeling and optimization of the active suspension system of a 6x6 electric vehicle

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ARTICLE INFO

Keywords:

Electric vehicle
In-wheel motor
Mathematical modeling
Genetic algorithm optimization
ISO 2631-1
Particle swarm optimization
Active suspension
Differential evolution optimization

ABSTRACT

Vehicle suspension systems have a significant role in the capacity of transmit forces coming from the road and in vehicle performance. Loads between the road and the vehicle body significantly affect driver health, comfort, and vehicle drive features. Some research shows that the quality of suspension systems is of great importance in vehicle accident rates. The suspension system also affects roadholding significantly. In this study, mathematical modeling was performed for a vehicle with a 6×6 in-wheel electric motor. A free body diagram was generated for the system and the wheels, vehicle body, seat, human model, and in-wheel electric motors were addressed in the mathematical simulation modeling. “Bump type” and “Random type” road model scenarios as road inputs were created. In the mathematical model, simulations of models with passive and active suspension systems were carried out and system accuracy was confirmed. Three different methods were used for optimization: Genetic Algorithm (GA), Particle Swarm Optimization (PSE) and Differential Evolution (DE) Optimization. The stiffness and damping coefficients of the wheel suspension, the coefficients of the active suspension controllers, the stiffness and damping coefficients of the seat suspension and the suspension of wheel-mounted motors are selected as design variables of the optimization problem. The optimizations were made separately and compared for two different road scenarios. According to the ISO 2631-1 Health Guide standard, it was evaluated which system is more efficient in terms of its effects on human health and comfort. ISO 2631-1, which defines vibration exposure limits, has been used to calculate driver health limits. Optimization results with active suspension systems showed better improvement compared with the simulations performed with passive suspension systems. Moreover, optimization results especially in systems used for a long time demonstrate greater results on human health.

1. Introduction

Vehicle suspension systems play a crucial role in damping the forces transmitted from the road and significantly contribute to overall vehicle performance. Furthermore, the interaction between the vehicle body and the road surface has a substantial impact on driver comfort and health [1]. Furthermore, suspension systems directly influence vehicle handling performance, road holding, and safety [2]. Various studies have shown that suspension systems are the determining factor in road holding and accident rates [3].

Vehicle suspension systems can be categorized into three types: passive, active, and semi-active suspension systems [1,4]. These systems are as critical in electric vehicles (EVs) as they are in traditional internal combustion engine (ICE) vehicles. In electric vehicles equipped with

in-wheel electric motors (IWMs), suspension systems present both opportunities for space optimization and challenges for suspension design [2]. Various studies focus on the analysis and optimization of suspension systems in electric vehicles, especially those with in-wheel motors [5–11]. GA, PSO, and DE were selected for suspension optimization due to their effectiveness in solving nonlinear, multi-parameter problems. Unlike gradient-based methods, which may get stuck in local minima, these population-based algorithms efficiently explore complex design spaces, making them well-suited for multi-modal optimization with multiple constraints.

In research involving various types of optimizations, different optimization methods are often compared. This study exemplifies such research. For instance, studies [12,13] serve as references for these comparisons. The paper [12] discusses the parametric synthesis of two

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Received 31 October 2024; Received in revised form 3 March 2025; Accepted 27 June 2025

Available online 11 July 2025

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Fig. 1. 6 × 6 Military vehicle [15].

different trunk lid mechanisms used in vehicles. Research [13] used a half-vehicle model and employed three different optimizations. The design and optimization of these mechanisms were performed using population-based optimization techniques such as Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Differential Evolution (DE) optimization. GA, PSO, and DE were chosen for suspension optimization due to their effectiveness in solving nonlinear, multi-parameter problems. Unlike gradient-based methods, which are prone to getting trapped in local minima, these population-based algorithms efficiently explore complex design spaces, making them highly suitable for multi-modal optimization with multiple constraints. The performance of each algorithm varied under different conditions; for instance, PSO demonstrated the highest efficiency in certain cases, whereas GA outperformed the others in different scenarios.

In this study, the active suspension system and controllers of the electric vehicle are optimized to meet the ride comfort and health criteria of the International Standardization Organization 2631–1 [14]. In order to do that, three optimization models are used: Genetic Algorithm (GA) optimization, Particle Swarm Optimization (PSO) and Differential Evolution (DE) Optimization. A comparison of these three optimization models was also conducted. It is observed that the active suspension system can significantly reduced the maximum amplitude of the vehicle acceleration and also the PSO is the best one to determine most successful vehicle parameters. To ensure statistical reliability, multiple simulations were conducted using a random road model, with at least 100 iterations for each optimization algorithm (GA, PSO, and DE). The mean and standard deviation of the results were analyzed to account for stochastic variability, thereby enhancing the robustness of the findings. This integrated approach highlights the potential of population-based optimization techniques in advancing suspension system design for electric vehicles, offering promising directions for future research in enhancing ride comfort and overall vehicle safety.

2. Mathematical modeling of 6 × 6 electrical vehicle

In this study, a 6 × 6 full vehicle model has been developed, and the vehicle's movements along various axes have been calculated. The model includes a vehicle with 15 degrees of freedom and a human model with 5 degrees of freedom. During the mathematical modeling process, the rotations and roll motions of the vehicle's body were specifically considered. This conducted 6 × 6 electrical vehicle can be used as a military vehicle such as [15]. The determined values were inspired from this conventional vehicle given in Fig. 1.

The vehicle model consists of 6 wheels and motors connected to these wheels. Vertical movement is allowed in these units, meaning the wheels and motors can move vertically. For the vehicle body, vertical movement, as well as pitch and roll angles, were included in the mathematical modeling process. The human model comprises a seat, pelvis system, lower torso, upper torso, and driver's head, amounting to

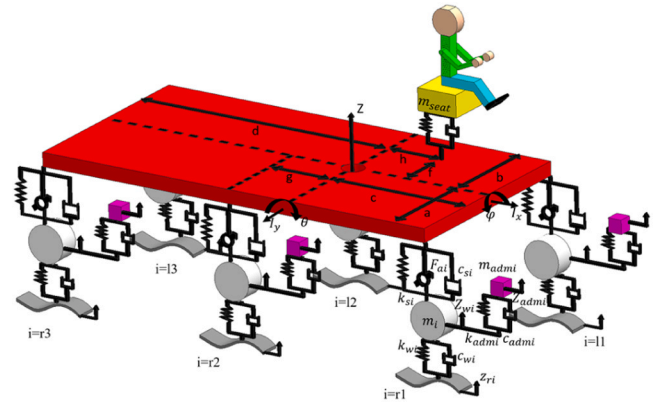


Fig. 2. 6 × 6 Electrical vehicle model.

5 spring systems.

The sprung systems in the vehicle model and the human model are connected to each other through suspension systems. These suspension systems consist of springs and damping elements.

Fig. 2 shows a version of the complete model. M , I_x , I_y , Z , Θ , and Φ represent the mass, pitch and roll moments of inertia, vertical movement, and pitch and roll angles of the vehicle body, respectively. m_i , k_{si} , c_{si} , F_{ai} , m_{admi} , k_{admi} , c_{admi} , k_{wi} , and c_{wi} respectively represent the wheel mass, spring constants, damping coefficients, mass of the electric motors, suspension spring constant of the electric motors, damping coefficient, spring, and damping constants of the wheel tire. $admi$ indicates the elements influenced by the internal electric motors. Z_{wi} and Z_{admi} denote the vertical displacements of the wheels and electric motors. a , b , c , d , g , h , f represents the system dimensions and specify the positions of the wheels and seats relative to the center of gravity. $i = r1, r2, r3, l1, l2, l3$ indicate the positions of the six wheels, respectively representing the front right, middle right, rear right, front left, middle left, and rear left wheel positions.

6 × 6 electric vehicle's Degrees of Freedom (DOF) and defines the vehicle model's force and as presented in Eqs. (1) to (6). These equations describe the dynamics of the vehicle, accounting for various forces and movements that the model undergoes.

$$F_{sr1} = k_{sr1} * (Z - c * \theta - a * \phi - Z_{wr1}) + c_{sr1} * (\dot{Z} - c * \dot{\theta} - a * \dot{\phi} - \dot{Z}_{wr1}) \quad (1)$$

$$F_{sr2} = k_{sr2} * (Z + g * \theta - a * \phi - Z_{wr2}) + c_{sr2} * (\dot{Z} + g * \dot{\theta} - a * \dot{\phi} - \dot{Z}_{wr2}) \quad (2)$$

$$F_{sr3} = k_{sr3} * (Z + d * \theta - a * \phi - Z_{wr3}) + c_{sr3} * (\dot{Z} + d * \dot{\theta} - a * \dot{\phi} - \dot{Z}_{wr3}) \quad (3)$$

$$F_{sl1} = k_{sl1} * (Z - c * \theta + b * \phi - Z_{sl1}) + c_{sl1} * (\dot{Z} - c * \dot{\theta} + b * \dot{\phi} - \dot{Z}_{sl1}) \quad (4)$$

$$F_{sl2} = k_{sl2} * (Z + g * \theta + b * \phi - Z_{sl2}) + c_{sl2} * (\dot{Z} + g * \dot{\theta} + b * \dot{\phi} - \dot{Z}_{sl2}) \quad (5)$$

$$F_{sl3} = k_{sl3} * (Z + d * \theta + b * \phi - Z_{sl3}) + c_{sl3} * (\dot{Z} + d * \dot{\theta} + b * \dot{\phi} - \dot{Z}_{sl3}) \quad (6)$$

Eqs. (7) to (12) describe the force between road and wheels,

$$F_{wr1} = k_{wr1} * (Z_{wr1} - Z_{rr1}) + c_{wr1} * (\dot{Z}_{wr1} - \dot{Z}_{rr1}) \quad (7)$$

$$F_{wr2} = k_{wr2} * (Z_{wr2} - Z_{rr2}) + c_{wr2} * (\dot{Z}_{wr2} - \dot{Z}_{rr2}) \quad (8)$$

$$F_{wr3} = k_{wr3} * (Z_{wr3} - Z_{rr3}) + c_{wr3} * (\dot{Z}_{wr3} - \dot{Z}_{rr3}) \quad (9)$$

$$F_{wl1} = k_{wl1} * (Z_{wl1} - Z_{rl1}) + c_{wl1} * (\dot{Z}_{wl1} - \dot{Z}_{rl1}) \quad (10)$$

$$F_{wl2} = k_{wl2} * (Z_{wl2} - Z_{rl2}) + c_{wl2} * (\dot{Z}_{wl2} - \dot{Z}_{rl2}) \quad (11)$$

$$F_{wl3} = k_{wl3} * (Z_{wl3} - Z_{rl3}) + c_{wl3} * (\dot{Z}_{wl3} - \dot{Z}_{rl3}) \quad (12)$$

Eqs. (13) and (18) represent the forces between the in-wheel electric motors and the wheels,

$$F_{admr1} = k_{admr1} * (Z_{admr1} - Z_{wr1}) + c_{admr1} * (\dot{Z}_{admr1} - \dot{Z}_{wr1}) \quad (13)$$

$$F_{admr2} = k_{admr2} * (Z_{admr2} - Z_{wr2}) + c_{admr2} * (\dot{Z}_{admr2} - \dot{Z}_{wr2}) \quad (14)$$

$$F_{admr3} = k_{admr3} * (Z_{admr3} - Z_{wr3}) + c_{admr3} * (\dot{Z}_{admr3} - \dot{Z}_{wr3}) \quad (15)$$

$$F_{adml1} = k_{adml1} * (Z_{adml1} - Z_{wl1}) + c_{adml1} * (\dot{Z}_{adml1} - \dot{Z}_{wl1}) \quad (16)$$

$$F_{adml2} = k_{adml2} * (Z_{adml2} - Z_{wl2}) + c_{adml2} * (\dot{Z}_{adml2} - \dot{Z}_{wl2}) \quad (17)$$

$$F_{adml3} = k_{adml3} * (Z_{adml3} - Z_{wl3}) + c_{adml3} * (\dot{Z}_{adml3} - \dot{Z}_{wl3}) \quad (18)$$

Eqs. (19) to (21) represent the motion equations of the vehicle body, which describe how the vehicle's center of mass moves in response to various forces and moments acting on it. These equations are crucial for understanding the vehicle's overall dynamics, including its vertical, pitch, and roll movements.

$$M\ddot{Z} = (-F_{sr1} - F_{sr2} - F_{sr3} - F_{sl1} - F_{sl2} - F_{sl3} - F_{ar1} - F_{ar2} - F_{ar3} - F_{al1} - F_{al2} - F_{al3} + F_{seat}) \quad (19)$$

$$I_y\ddot{\theta} = ((F_{sr1} + F_{sl1} + F_{ar1} + F_{al1}) * c - (F_{sr2} + F_{sl2} + F_{ar2} + F_{al2}) * g - (F_{sr3} + F_{sl3} + F_{ar3} + F_{al3}) * d - F_{seat} * f) \quad (20)$$

$$I_x\ddot{\phi} = ((F_{sr1} + F_{sr2} + F_{sr3} + F_{ar1} + F_{ar2} + F_{ar3}) * a - (F_{sl1} + F_{sl2} + F_{sl3} + F_{al1} + F_{al2} + F_{al3}) * b + F_{seat} * h) \quad (21)$$

Eqs. (22) to (27) describe the motion dynamics of the wheels in a vehicle model. These equations capture the interactions between the wheels and the road surface, including forces,

$$M_{wr1}\ddot{Z}_{wr1} = (F_{sr1} - F_{wr1} + F_{admr1} + F_{ar1}) \quad (22)$$

$$M_{wr2}\ddot{Z}_{wr2} = (F_{sr2} - F_{wr2} + F_{admr2} + F_{ar2}) \quad (23)$$

$$M_{wr3}\ddot{Z}_{wr3} = (F_{sr3} - F_{wr3} + F_{admr3} + F_{ar3}) \quad (24)$$

$$M_{wl1}\ddot{Z}_{wl1} = (F_{sl1} - F_{wl1} + F_{adml1} + F_{al1}) \quad (25)$$

$$M_{wl2}\ddot{Z}_{wl2} = (F_{sl2} - F_{wl2} + F_{adml2} + F_{al2}) \quad (26)$$

$$M_{wl3}\ddot{Z}_{wl3} = (F_{sl3} - F_{wl3} + F_{adml3} + F_{al3}) \quad (27)$$

Eqs. (28) - (33) show In-Wheel Electric Motor Motion Equations,

$$M_{admr1}\ddot{Z}_{admr1} = -F_{admr1} \quad (28)$$

$$M_{admr2}\ddot{Z}_{admr2} = -F_{admr2} \quad (29)$$

$$M_{admr3}\ddot{Z}_{admr3} = -F_{admr3} \quad (30)$$

$$M_{adml1}\ddot{Z}_{adml1} = -F_{adml1} \quad (31)$$

$$M_{adml2}\ddot{Z}_{adml2} = -F_{adml2} \quad (32)$$

$$M_{adml3}\ddot{Z}_{adml3} = -F_{adml3} \quad (33)$$

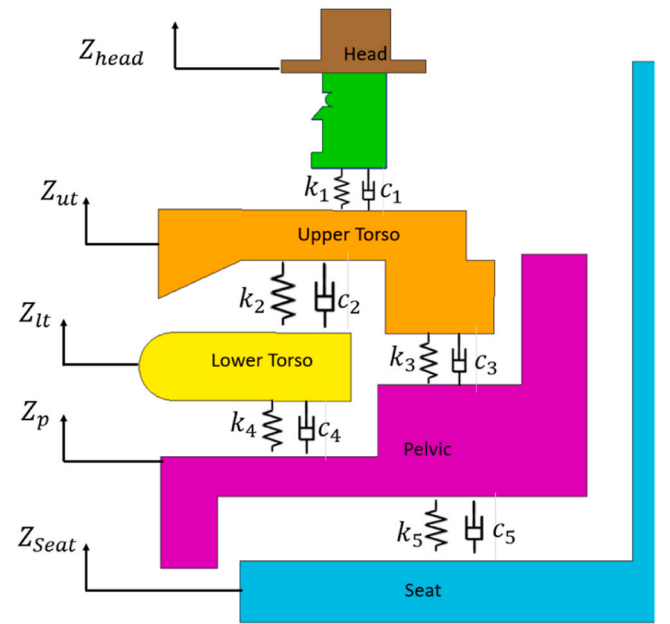


Fig. 3. Biodynamic human model.

The biodynamic human model, as illustrated in Fig. 3 and system created by using NX software. This model is used to simulate and analyze the biomechanical behavior of the human body under various dynamic conditions. It can be used in applications such as automotive safety, ergonomics, and health diagnostics to study the impact of forces and motions on the human body.

Eqs. (34) to (35) describe the forces and motions associated with the vehicle seat, which is an integral part of the biodynamic human model.

$$F_{seat} = k_{seat} * (Z_{seat} - (Z - f * \theta + h * \phi)) + c_{seat} * (\dot{Z}_{seat} - (\dot{Z} - f * \dot{\theta} + h * \dot{\phi})) \quad (34)$$

$$M_{seat}\ddot{Z}_{seat} = F_p - F_{seat} \quad (35)$$

Eqs. (36) to (40) show the forces acting on different parts of the human body model within the biodynamic simulation,

$$F_p = k_5 * (Z_p - Z_{seat}) + c_5 * (\dot{Z}_p - \dot{Z}_{seat}) \quad (36)$$

$$F_{lp} = k_4 * (Z_{lt} - Z_p) + c_4 * (\dot{Z}_{lt} - \dot{Z}_p) \quad (37)$$

$$F_{up} = k_3 * (Z_{ut} - Z_p) + c_3 * (\dot{Z}_{ut} - \dot{Z}_p) \quad (38)$$

$$F_{ult} = k_2 * (Z_{ut} - Z_{lt}) + c_2 * (\dot{Z}_{ut} - \dot{Z}_{lt}) \quad (39)$$

$$F_h = k_1 * (Z_h - Z_{ut}) + c_1 * (\dot{Z}_h - \dot{Z}_{ut}) \quad (40)$$

Eqs. (41) to (44) represent the motion of the human body model. They account for the forces and accelerations acting on different parts of the human body, contributing to a comprehensive understanding of the biodynamic response.

$$M_p\ddot{Z}_p = (F_{up} + F_{lp} - F_p) \quad (41)$$

$$M_{lt}\ddot{Z}_{lt} = (-F_{lp} + F_{ult}) \quad (42)$$

$$M_{ut}\ddot{Z}_{ut} = (-F_{ult} - F_{up} + F_h) \quad (43)$$

$$M_h\ddot{Z}_h = (-F_h) \quad (44)$$

Table 1
For the bump model road type, the variables.

Variable	Value	Unit
a	0.5	(m)
Lamda	60	(m)
V	60	(km/h)
L _{front}	2.2	(m)
L _{mid}	0.001	(m)
L _{rare}	2.2	(m)
SS _{mid}	2.201	(m)
SS _{rare}	4.4	(m)

Table 2
For the random model road type, the variables [15].

Variable	Value	Unit
ρ	0.45	(m ⁻¹)
σ ²	300	(mm ²)
V	60	(km/h)

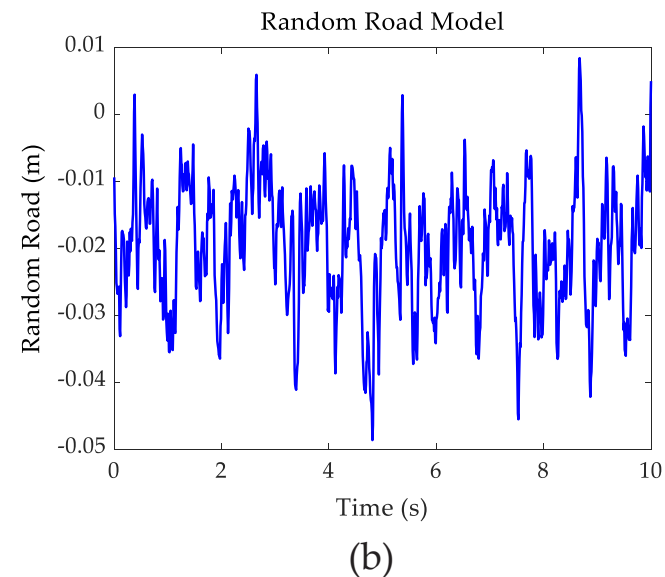
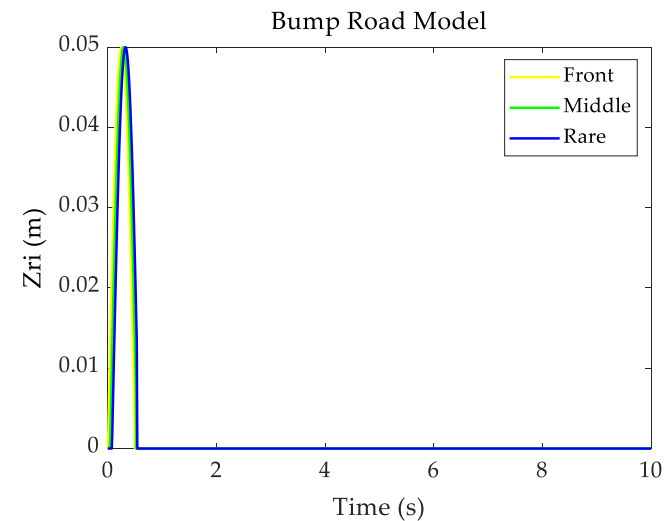


Fig. 4. (a) Bump type and (b) road model.

Table 3
Settings used for the genetic algorithm [15], PSO and DE.

Variable	Value
Population size	500
No of generations	100
Fitness scaling	Rank
Mutation technique	Mutation uniform

During the simulation, two different types of road models are used: bump model and random road model. The Eqs. (45)-(48) related to the Bump type road models are given below, and the constants are provided in Table 1. The road modeling is based on the work by [14,16].

$$Z_{r,1l} = \begin{cases} a^* \sin(w^*t) & 0 \leq t \leq a^* \frac{\text{lamda}}{v} \\ \text{Odiğer durumlar} & \end{cases} \quad (45)$$

$$Z_{r,12} = \begin{cases} a^* \sin(w^*(t - \text{ssmid})) & \text{ssmid} \leq t \leq a^* \left(\text{ssmid} + \frac{\text{lamda}}{v} \right) \\ \text{Odiğer durumlar} & \end{cases} \quad (46)$$

$$Z_{r,12} = \begin{cases} a^* \sin(w^*(t - \text{ssrare})) & \text{ssrare} \leq t \leq a^* \left(\text{ssrare} + \frac{\text{lamda}}{v} \right) \\ \text{Odiğer durumlar} & \end{cases} \quad (47)$$

$$W = (2 * \pi * V / \text{lamda}) \quad (48)$$

The second road profile is the random road model, created using Eqs. (49)-(50). It is simulated using white noise deviation [15]. The data for the random road model is provided in Table 2.

$$\dot{Z}_{r,li} + \rho * V * Z_{r,li} = V * w_n \quad (49)$$

$$w_n = 2 * \sigma^2 * \rho * V \quad (50)$$

The road models showing in Fig. 4 have been used in the system modeling. The reason for the different fluctuations in the bump-type road model is due to the vehicle entering the bump at different times. The responses of the wheels to the road's feedback occurred at different times.

In this study, three optimization systems are used. First, variable suspension parameters are adjusted to achieve the most efficient vehicle suspension performance and the best driver comfort. These parameters include the suspension systems between the wheel and the body, the suspension systems for the electric motor, the seat suspension, and the active suspension system. These variables are selected for optimization.

The optimization conditions and the lower and upper limits of the

Table 4
Data to be used for DE, PSO and genetic algorithm upper and lower limit [15].

Variable	Value	Unit
Nvar	28	-
K _{si}	Upper Bond: 800000 Lower Bond: 200000	N/m
C _{si}	Upper Bond: 53000 Lower Bond: 13000	N/(m/s)
K _{admi}	Upper Bond: 1500000 Lower Bond: 800000	N/m
C _{admi}	Upper Bond: 30000 Lower Bond: 15000	N/(m/s)
K _{seat}	Upper Bond: 30000 Lower Bond: 7500	N/m
C _{seat}	Upper Bond: 1400 Lower Bond: 350	N/(m/s)
K _p	Upper Bond: 150000 Lower Bond: 50000	N/m
K _d	Upper Bond: 1500 Lower Bond: 0	N/(m/s)

suspension system variables used in the modeling are shown in Table 3, which includes the GA, PSO and DE algorithm conditions.

The upper and lower limits of the vehicle design parameters are selected by considering specific factors. The variable ranges are provided in Table 4. The design variables include the main suspension system of the vehicle (stiffness and damping coefficients), the driver seat suspension system (stiffness and damping coefficients), and the cushion stiffness and damping coefficient of the driver seat.

The mathematical models and values provided above have been implemented using the methods described below

2.1. Runge-Kutta mathematical solving method

In this study, the simulation of a 6×6 electric vehicle along with a human model was conducted, and the free body diagram of the model was developed. The vehicle model has 15 degrees of freedom, while the driver model, including the seat, has 5 degrees of freedom. The study examined the responses of passive, active, and optimized suspension systems to bump and random road profiles, with a focus on their effects on driver comfort. The results were analyzed to determine the most suitable conditions for optimal performance. Mathematical modeling was employed for the system simulation.

The Runge-Kutta method is widely used to solve mathematical problems involving differential equations and is particularly suitable for first-order differential equations. This method, frequently utilized in MATLAB and other scientific computing tools, calculates the evolution of differential equations over time.

In our system, 'ode45' solver was utilized. The solution process begins by defining the initial conditions and the constants involved in the mathematical model. The time span for the solution, referred to as 'tspan', must be specified within the system. The model is capable of performing calculations using coefficients and solving derivative functions. Advantages of this calculation system include high accuracy, usability in most scientific computation software, and a wide range of applications for various differential equations. However, disadvantages include potential errors and accuracy deviations when a large step size is chosen. According to the detailed scientific computation guidebook, the Runge-Kutta method appears suitable for the simulation [17]. In this research and the setup of the mathematical model, optimization research by [14] and the computer-aided engineering textbook were used [16].

This book is a classic reference source where scientific computation methods are discussed in detail. It explains numerous numerical methods, including the Runge-Kutta method [17].

2.2. PD controller system

In the field of active suspension control, researchers have investigated various methods to improve ride comfort and handling performance [5,8]. Given the conflicting nature of performance criteria such as comfort, handling, and suspension displacement, numerous control strategies have been explored. These include Linear Quadratic Regulator (LQR) [18], Adaptive Sliding Control [19], H ∞ control [20], Sliding Mode Control [21], Fuzzy Logic [7], Proportional-Integral (PI) Control [22], and Neural Networks [23], each addressing different trade-offs in active suspension systems. However, due to their complexity and high computational demands, some of these approaches may not perform efficiently in real-time applications or unfamiliar operating conditions.

Among these methods, Proportional-Derivative (PD) controllers are commonly utilized in industrial applications due to their simplicity and reliable performance. A PD controller generates control signals by minimizing the error through a feedback loop, relying on two key gain parameters:

Proportional Gain (K_p): Determines the control force based on the present error, applying a correction proportional to the deviation from the target position.

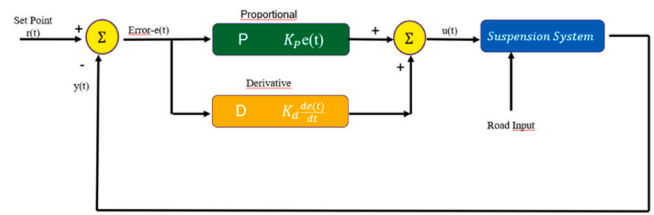


Fig. 5. Loop PD controller.

Derivative Gain (K_d): Responds to the rate of change of the error, enhancing system stability by predicting future deviations and mitigating oscillations.

The PD controller loop shown in Fig. 5 and follows a specific control law, which governs its behavior in regulating system performance shown in Eq. (51).

$$u(t) = K_p e(t) + K_d \frac{d_e(t)}{dt} \quad (51)$$

$u(t)$ is the control signal.

- $e(t)$ is the error (difference between the desired and actual position),
- $\frac{d_e(t)}{dt}$ is the derivation of the error, Providing damping to the system [19].

For systems operating under varying conditions, PD controllers provide an efficient and straightforward control mechanism. By adjusting the proportional and derivative gains, the system response can be fine-tuned. However, in highly nonlinear and dynamic environments, PD controllers alone may not always achieve optimal performance [24, 25].

The overall efficiency of a PD controller is heavily dependent on the tuning of its gain parameters. Traditional tuning techniques, such as trial-and-error, Ziegler–Nichols, and gradient descent, are available; however, these methods can be time-intensive and may not yield optimal results, particularly in complex multi-objective optimization problems. In this study, the optimization process focused on determining the most effective values for proportional gain (K_p) and derivative gain (K_d) [26].

2.3. GA optimization

Optimization of vehicle systems is of considerable importance in automotive engineering. The performance of suspension systems directly affects crucial areas such as road handling, driving comfort, and vehicle dynamics. Many complex mathematical models are developed in the automotive field, and optimizations are performed to determine the best working model of the system by specifying boundary conditions. Universal computational methods such as Genetic Algorithms (GA) are frequently used for complex mathematical optimization systems. There are various reasons why genetic algorithms are preferred for the simulation and optimization of vehicle suspension systems.

Mathematical modeling of automotive suspension systems is classified as a complex problem. Suspension systems comprise numerous parameters and constraints, and GA systems are effectively utilized in multidimensional search spaces [27,28]. Optimization can be carried out according to different performance criteria in complex spatial systems. Considering their potential to achieve global optimization, genetic algorithms are more successful than local optimization methods, which operate within a very limited area. Therefore, global optimization methods are more suitable for this system. PSO is also a type of global optimization method [29].

GA optimization systems provide flexibility in computations. The system is optimized by defining objective functions. Separate optimizations can be performed for the driver and passengers in the rear of the

vehicle [30].

Genetic algorithms can exhibit robust performance even in noisy and uncertain problem environments [31,32]. This robustness is highly valuable for systems like suspension systems that encounter various road conditions and dynamic loads. Genetic algorithms can produce consistent and reliable solutions even in environments with uncertainty and variability. Because of these features, they stand out as an ideal method for optimizing the performance of vehicle suspension systems.

Genetic algorithms (GA) can incorporate boundary conditions into the mathematical modeling by defining variables and can perform optimization under various external conditions [33]. The functioning of a genetic algorithm requires the definition of its objective function. In our case, this function should include the parameters used in the optimization of the suspension system, along with the objective function for minimizing these parameters. Additionally, the parameters of the optimization system, such as population size and crossover rate, should be defined. Twenty-eight variable parameters need to be designed, with specified lower and upper limits for each parameter, which constrain the range within which the genetic algorithm can vary these parameters. The data for the variables, conditions, and parameters are sourced from [15].

When the solution process starts, each individual in the population represents a potential solution. The fitness value of each unit is calculated using the fitness function. The most suitable values are selected for crossover and mutation processes, providing better solutions. Crossovers provide diversity among individuals. The mutation system introduces changes to the units, which helps in identifying variables without getting lost in space and avoiding local maxima. Subsequently, new population individuals are formed, added to the population, and included in the generation. This process is repeated until the maximum number of iterations is reached.

The iterative and biologically inspired nature of GA ensures better solutions over time, and the flexibility of the algorithm allows it to be effectively applied to various optimization problems.

2.4. PSO optimization

Particle Swarm Optimization (PSO) is an effectively used method in optimization of vehicle suspension systems. This optimization technique is inspired by the collective behavior observed in bird flocks and fish schools [34]. The method establishes a swarm of particles, each representing potential optimal solutions within the search space. These particles continuously update their positions, with each particle characterized by a position and a velocity vector. While the position indicates a candidate solution, the velocity determines how the particle moves from one position to another [35].

Fitness functions are used to assess the performance of suspension systems in terms of their effectiveness and efficiency. These functions measure the degree to which specific targets are achieved. For vehicle suspension systems, fitness functions typically include road handling, driving comfort, durability, and fuel efficiency. In this study, the main function is driving comfort.

The PSO optimization process follows a series of steps. Initially, a population is generated with random combinations, and random particle swarms are defined. Each particle represents the necessary parameters for the suspension system [36]. Fitness evaluations are calculated according to the performance criteria of the suspension system. Particle velocities are updated based on their personal best positions (pBest) and the global best position of the swarm (gBest), and positions are recalculated based on the updated velocities. In each iteration, the best positions of the particles and the global position of the swarm are updated. These steps are repeated until a certain number of iterations or a fitness threshold is reached. At the end of the optimization process, the best position identified in the global system represents the optimal parameters for the suspension systems, ensuring maximum performance and efficiency. These parameters are adjusted to maximize system

performance and driving comfort [34].

One of the biggest advantages of PSO optimization is its fast convergence and ease of implementation, making it a practical solution for the optimization of suspension systems [35].

The PSO process begins with defining the initial position for each particle. The swarm size (ssize) and the number of variables (nvar) are defined, with lower and upper limits set for each variable. The initial positions and velocities of the particles within the swarm are also defined. The fitness value is calculated for the initial position of each particle, and personal best (pBest) and global best (gBest) positions are initially determined based on these values. Until the number of iterations or stopping criteria are met, the inertia weight (w) is updated in each iteration. Inertia weight determines the tendency to maintain the previous velocities of the particles and typically changes from high to low. Each particle's velocity is updated considering the personal best position (pBestpos) and the global best position (gBestpos). Each particle's velocity is kept within specified maximum and minimum velocity limits. Each particle's position is updated based on the updated velocities. The fitness value is calculated for each updated position. If the fitness criteria are met, the values are recorded; otherwise, they are not recorded, and the next iteration is proceeded. When the PSO cycle is completed, the best objective function value (gBestval) and the best solution (gBestpos) are displayed.

The PSO optimization process starts with the initialization step, where the initial population and velocities are determined, and initial fitness values are calculated. The PSO cycle begins with the random updating of positions [34]. During the cycle, inertia weights and velocities are updated, leading to changes in particle positions. Fitness values are recalculated based on these new positions, and personal and global values are updated to reflect the best positions [36]. The best fitness value and solutions are determined as the final results. This cycle illustrates the fundamental operation of the PSO algorithm and its application in optimizing vehicle suspension systems. The iterative nature of PSO allows for improved solutions over time, and the flexibility of the algorithm makes it suitable for various optimization problems [34,35].

PSO (Particle Swarm Optimization) has been used in some studies due to its simplicity and effectiveness in handling non-linear and multi-objective problems. For example, in study [37], PSO was chosen for the optimization of a non-linear half-vehicle suspension model. The data obtained from the study yielded highly successful results. vehicle vibrations were analyzed and minimized using a five-degree-of-freedom half-vehicle model. Optimization performed using the PSO algorithm significantly reduced the amplitudes of vehicle vibrations. For future studies, it is planned to validate these results through experiments and to integrate them with different controllers for active suspension systems [37].

In setting up the optimization system, support was provided by [16] and an open-source code was used.

2.5. Differential evolution optimization

Differential Evolution (DE) optimization is a population-based search algorithm that is often employed in solving complex optimization problems. DE is particularly effective in engineering applications, where it can be used to optimize parameters that are critical to system performance. In the context of vehicle suspension systems, which are key components influencing ride comfort and handling performance, optimized parameters are of paramount importance.

Reasons for preference: DE performs well in complex and multidimensional optimization problems. Optimizing the dynamic properties of suspension systems requires high accuracy, and DE can meet this requirement [38]. DE is prone to detect the global optimum solution without getting stuck in local minima. The optimization of suspension systems often involves multiple local minima, and DE is effective in such problems [39]. With its adjustable parameters (mutation factor and

crossover rate), DE allows for the development of suitable optimization strategies for various suspension systems and road conditions [29].

Briefly describing the optimization flow: An initial random population is generated within the solution space of the problem to explore potential solutions. Each individual represents a candidate solution, and the population size is generally determined based on the dimension of the solution space. For each target vector, three different vectors are randomly selected. The mutation vector is calculated. Crossover is performed between the target vector and the mutation vector. This allows for the creation of a trial vector for each target vector. Selection is made between the target vector and the trial vector. The vector that performs better according to the fitness function remains in the population. The process is repeated until the maximum number of iterations or an acceptable fitness level is reached [38]. Also, Fig. 3 shows optimization flow.

It has been observed that Differential Evolution (DE) provides good results in suspension systems. In the study of an article, bump model optimizations were performed on a half-vehicle system. Comparisons were made between PSO, GA, and DE optimizations, and the efficiency of DE optimization in the half-vehicle model was observed [13].

2.6. Evaluation of driver comfort and health with vibration

The ISO 2631–1:1997 standard, called "Mechanical Vibration and Shock – Evaluation of Human Exposure to Whole-body Vibration – Part 1: General Requirements," evaluates human health and comfort. It is used to minimize the health impacts of vibration exposure for workers and drivers [14,40]. This standard is also employed to assess passenger and driver comfort in vehicles and to develop strategies to reduce the effects of vibration in vehicle, machinery, and structure designs. It is widely used as a reference in scientific research.

In this study, the vibration acceleration method was used to evaluate vibration exposure. Mathematical calculations were performed using the Runge-Kutta method to determine the system's motion acceleration. A random road model was utilized to consider vibration exposure as a suitability function. Parameters optimized for the driver's health based on the duration of vibration exposure were identified, and the conditions under which these parameters were most effective were evaluated.

The Root Mean Square (RMS) acceleration method was chosen as the objective function in the research. RMS acceleration values were calculated based on the accelerations of the body, seat, and head of the human model within the system, and these values were used to evaluate the vibration levels in the system.

ISO standards defined the exposure limit which is determined by RMS weighted head acceleration. This criterion classifies vibration exposure into three categories: no health risk (NHR), potential health risk (PHR), and likely health risk (LHR) [14].

When defining the objective function, it is desired to minimize the motion accelerations for the body, seat, and human model. The error definition is made using the Root Mean Square (RMS). It is a method used to evaluate the heights of the wavelengths. The details are shared in Eq. (52).

The fundamental principle of this calculation method involves taking the mean of the squares of the focused parameters and then calculating the square root. This computation produces an average error value. This can be done separately for the seat, body, and human model optimizations, but the objective in this study is to achieve multi-objective optimization, as formulated in Eq. (52). It is an objective function prepared based on research examples [14,15].

Table 5

Vehicle model constants [15].

Paremater	Value	Unit
a	1.5	m
b	1.5	m
c	2.2	m
d	2.2	m
g	0.001	m
h	0.7	m
f	2.2	m
I _x	280000	kgm ²
I _y	2800	kgm ²
M	26800	kg
M _{wr1} , M _{wr2} , M _{wr3} , M _{wl1} , M _{wl2} , M _{wl3}	200	kg
M _{admr1} , M _{admr2} , M _{admr3} , M _{adml1} , M _{adml2} , M _{adml3}	500	kg
M _{seat}	30	kg
K _{wri}	1200000	N/m
C _{wi}	10	N/(m/s)

Table 6

Human model constants [41].

Paremater	Value	Unit
k ₁	166900	N/m
k ₂	10000	N/m
k ₃	144000	N/m
k ₄	20000	N/m
k ₅	49340	N/m
c ₁	310	N/(m/s)
c ₂	200	N/(m/s)
c ₃	909.1	N/(m/s)
c ₄	330	N/(m/s)
c ₅	2475	N/(m/s)
m _p	4.17	kg
m _{lt}	15	kg
m _{ut}	5.5	kg
m _h	36	kg

$$\text{Obj} = \sqrt{\left\{ \frac{1}{T} \int_0^T (a_{\text{BODY}}(t)/3)^2 dt \right\}} + \sqrt{\left\{ \frac{1}{T} \int_0^T (a_{\text{SEAT}}(t)/3)^2 dt \right\}} + \sqrt{\left\{ \frac{1}{T} \int_0^T (a_{\text{HEAD}}(t)/3)^2 dt \right\}} \quad (52)$$

PSO optimization will be used to compare the accuracy of optimization. This loop includes steps such as updating the velocities and positions of the particles, calculating fitness values, and determining the best solutions. This loop demonstrates the basic functioning of the PSO algorithm and how it can be used for optimizing the vehicle suspension system. The iterative nature of PSO allows for achieving better solutions over time, and the flexibility of the algorithm enables its effective application in various optimization problems. The lower and upper limit boundaries are provided in Table 4.

After the optimizations are performed, vibration exposure calculations are carried out according to ISO 2631–1. In these calculations, the objective function used is the RMS of the head movement acceleration in the human model. The function equation is shown in Eq. (53) below [12].

$$\text{RMS}_{a_{\text{head}}} = \sqrt{\left\{ \frac{1}{T} \int_0^T (a_{\text{HEAD}}(t))^2 dt \right\}} \quad (53)$$

Tables 5 and 6 contains the constant data that is used in the mathematical modeling. The calculation parameters related to human bio-dynamics, provided in Fig. 7, were adopted from the study by [15].

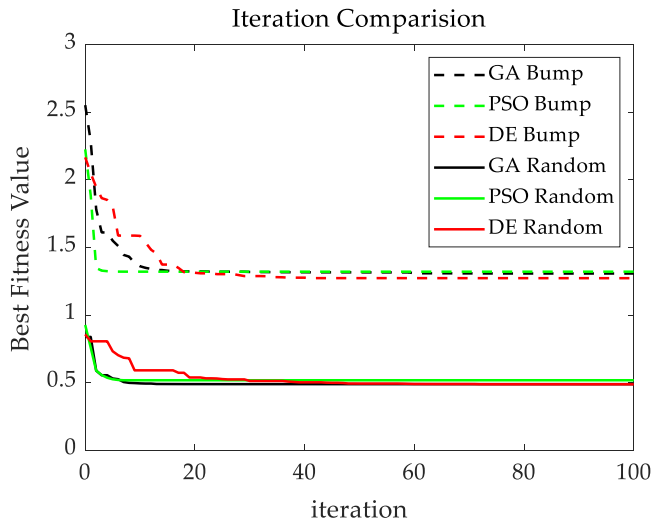


Fig. 6. Fitness function comparison.

The parameters presented in Table 6 are constant parameters to be used in the simulation of the human modeling.

The data in Table 5 are constant parameters that are used in the simulation of the vehicle suspension systems. These parameters will not be used during the optimization process.

3. Results

The simulation was conducted on a system with 15 degrees of freedom. There is a 5 degrees of freedom seat and human model. The model system was solved using MATLAB.

Initial parameters used for passive and active systems. Bump model Random Road Parameters on optimization determinate after

optimization and these parameters will remain unchanged.

Fig. 6 compares the fitness function in Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE) optimization processes. The characteristic of the following iterations is similar.

The results of the optimizations conducted under different scenarios are presented in Table 7. These scenarios include separate optimizations for the main suspension system, the driver seat suspension system, and the combined system.

3.1. Passive and active system comparison

The comparison focuses on the vehicle's passive and active suspension systems and the simulations performed on bump-type and random road types. The analyses will determine which system is more efficient.

3.1.1. Bump type road model passive and active comparison

The bump road model simulates a road with bumps, designed to test the suspension system's response to uneven surfaces. The timing of the wheels encountering the bumps differs, making the simulation more accurate.

As shown in Fig. 7, when comparing the harmonic motions for the vehicle body, seat, and head displacements, it highlights how road feedback affects various parts of the system as the vehicle traverses a bumpy road. Fig. 8 shows the acceleration movements of harmonic motions occurring in specific areas of the vehicle system.

3.1.2. Random type road model passive and active comparison

The random road model simulates traveling on a rough road. When comparing the results, the changes in the vehicle body, seat and head of human system are considered.

As shown in Fig. 9, the first stage of the comparison of optimizations for the random road model is based on the vertical displacement movements of the body, seat, and the head of the human model, and in Fig. 10, it is based on acceleration movements. As shown on the graphs,

Table 7
Suspension values before and after optimization.

Variables	Initial	Bump Road model			Random Road Model			Unit
		GA	PSO	DE	GA	PSO	DE	
K _{sr1}	400000	200000	200000	200000	200000	200000	200000	N/m
K _{sr2}	400000	200000	200000	200000	200000	200000	200000	N/m
K _{sr3}	400000	211554.486	200000	200000	200000	200000	200000	N/m
K _{sl1}	400000	200000	200000	200000	200000	200000	200000	N/m
K _{sl2}	400000	200000	800000	200000	200000	200000	200000	N/m
K _{sl3}	400000	200000	800000	200000	200000	277510.6	200000	N/m
C _{sr1}	26667	14187.504	13000	13000.2	13000	13000	13000	N/(m/s)
C _{sr2}	26667	25840.334	13000	28426.71	13000	53000	13000	N/(m/s)
C _{sr3}	26667	52856.789	53000	53000	48401.30	53000	46162.56	N/(m/s)
C _{sl1}	26667	45664.631	50909.3	43785.97	13000	13000	13000	N/(m/s)
C _{sl2}	26667	13000	53000	13000	13000	13000	13000	N/(m/s)
C _{sl3}	26667	507644.152	53000	53000	50764.42	53000	53000	N/(m/s)
K _{admr1}	200000	800000	800000	800000	800000	800000	800000	N/m
K _{admr2}	200000	860347.636	800000	800000	800000	1500000	808201.83	N/m
K _{admr3}	200000	1293136.60	800000	1220371.1	800000	1500000	1110371.10	N/m
K _{adml1}	200000	868274.55	1500000	940000	800000	1500000	800000	N/m
K _{adml2}	200000	800000	1500000	800000	800000	800000	800000	N/m
K _{adml3}	200000	1104567.13	800000	1476994.9	808323.47	1500000	1039804.47	N/m
C _{admr1}	40000	27000.4366	29705.1	30000	15029.235	15000	15000	N/(m/s)
C _{admr2}	40000	16985.8303	15000	20743.06	15000	30000	15000	N/(m/s)
C _{admr3}	40000	21604.501	30000	28687.4	29935.463	30000	27000	N/(m/s)
C _{adml1}	40000	28501.986	30000	30000	15000	15000	15000	N/(m/s)
C _{adml2}	40000	24597.572	15000	22088.79	15000	15000	15000	N/(m/s)
C _{adml3}	40000	17692.834	30000	15000	29934.726	15000	26099.725	N/(m/s)
K _{seat}	15000	7500	7500	200000	7500	7500	7500	N/m
C _{seat}	700	1398.877	1400	200000	1285.9785	1161.07	1300.45	N/(m/s)
K _p	Passive:0 Aktif:150000	149223.18	150000	150000	149279.55	150000	150000	N/m
K _d	Passive:0 Aktif: 1500	1496.722	1500	500	500	500	500	N/(m/s)

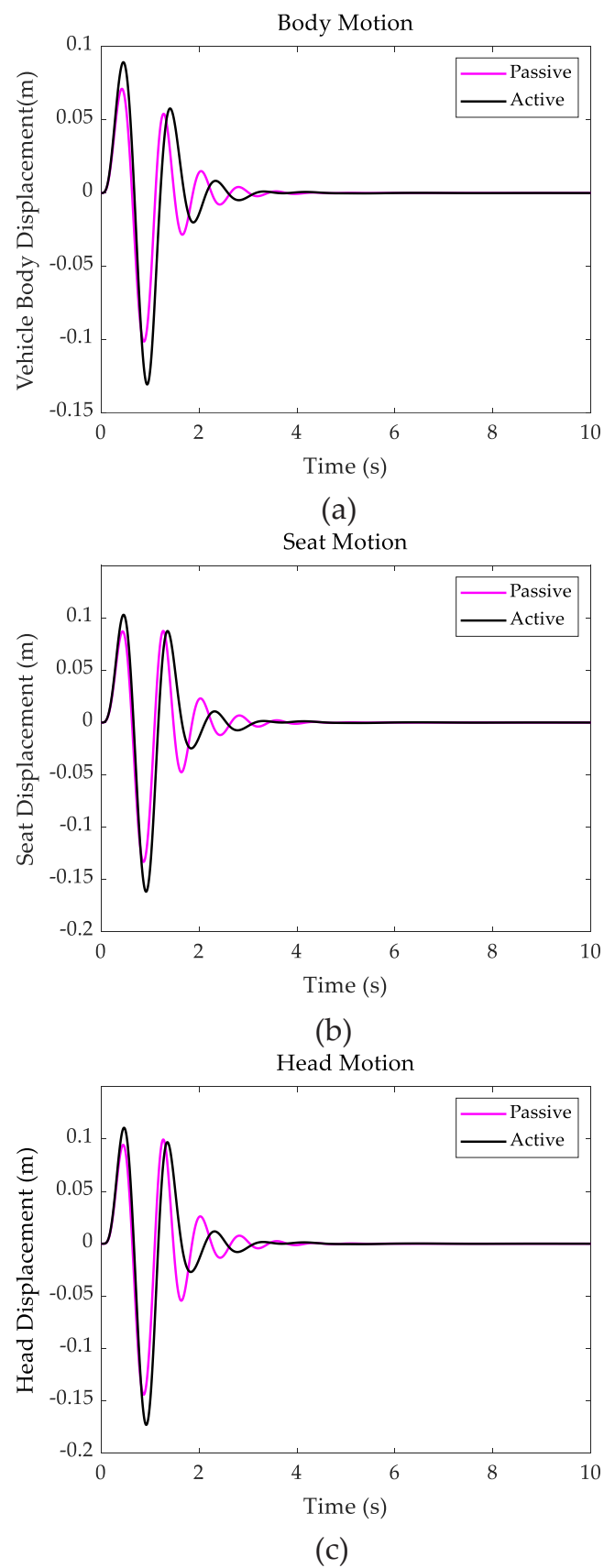


Fig. 7. The vehicle dynamics on a bump road model: (a) Body vertical displacement, (b) Seat vertical displacement, (c) Head displacement of human body.

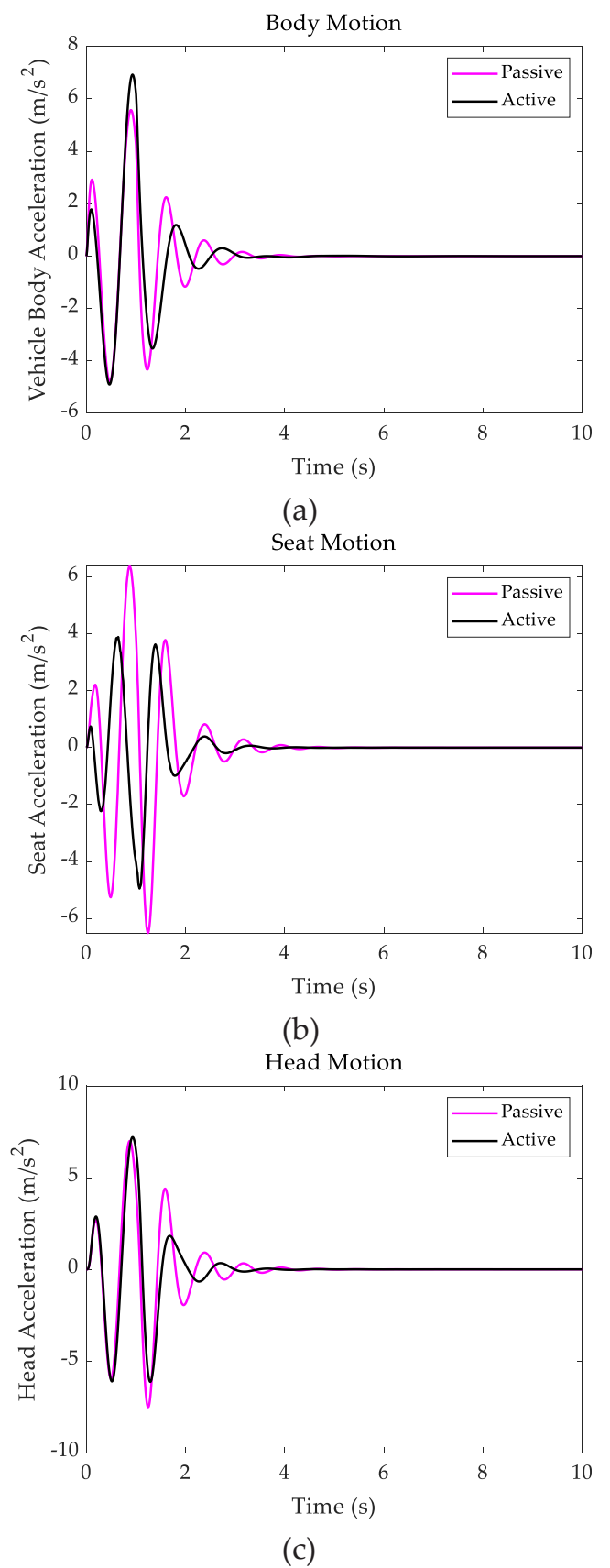
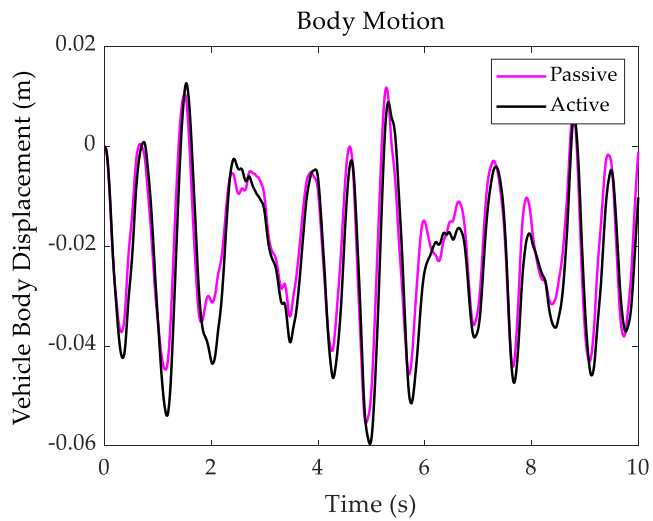
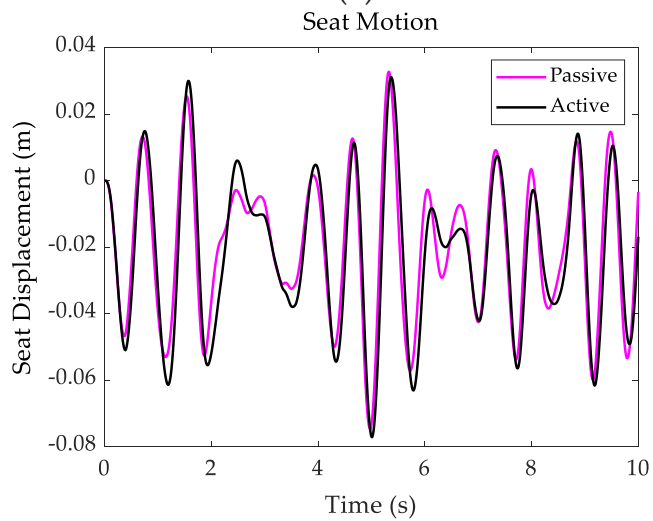


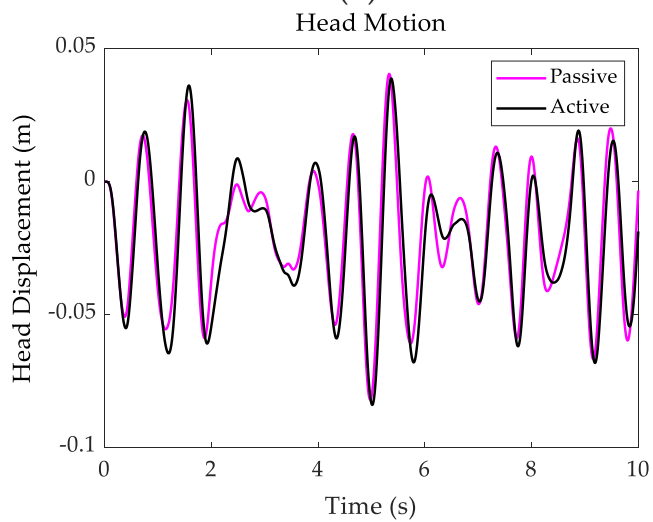
Fig. 8. The vehicle dynamics on a bump road are analyzed (a) Body vertical acceleration, (b) Seat vertical acceleration, (c) Head acceleration of Human Body.



(a)

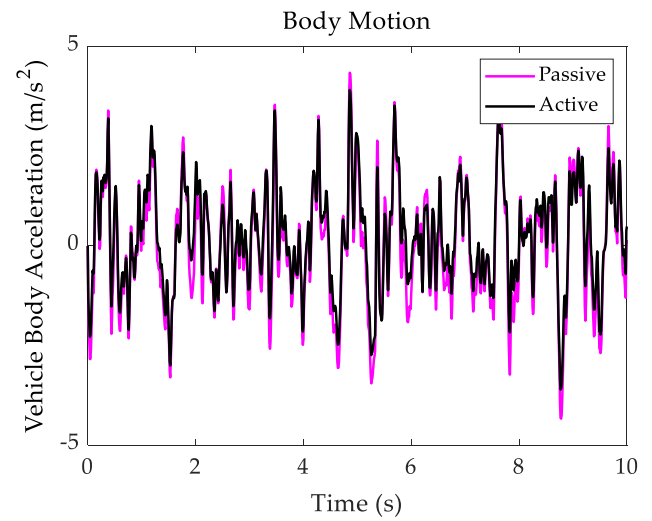


(b)

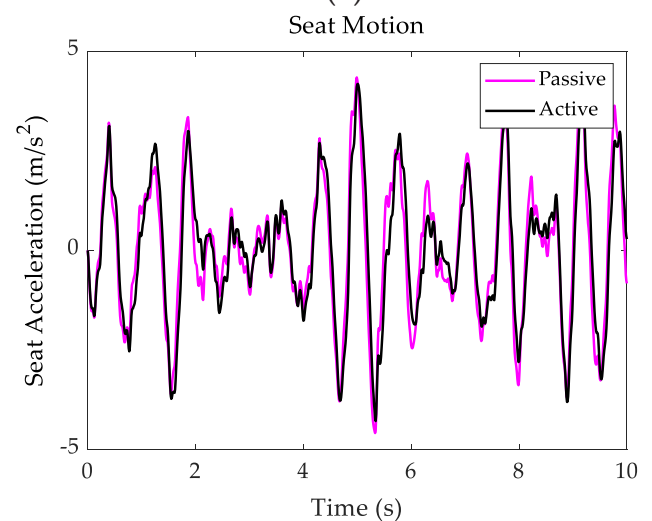


(c)

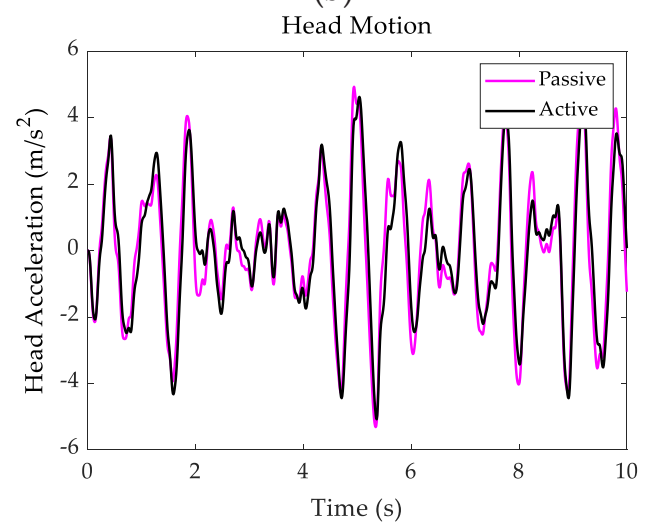
Fig. 9. The vehicle dynamics on a random road are analyzed (a) Body displacement, (b) Seat displacement, (c) Head displacement of human body.



(a)

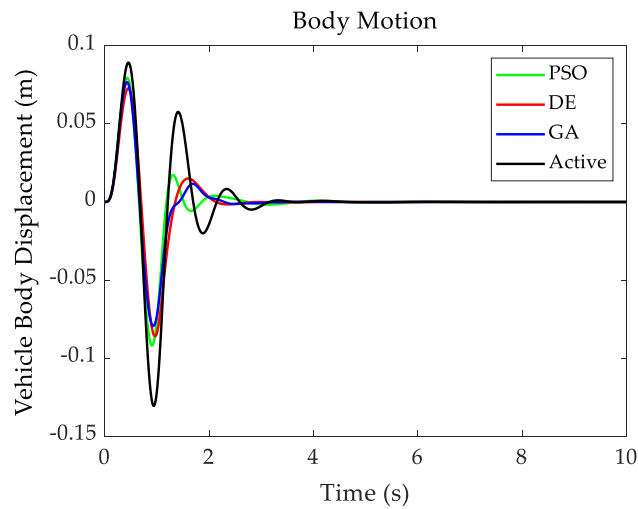


(b)

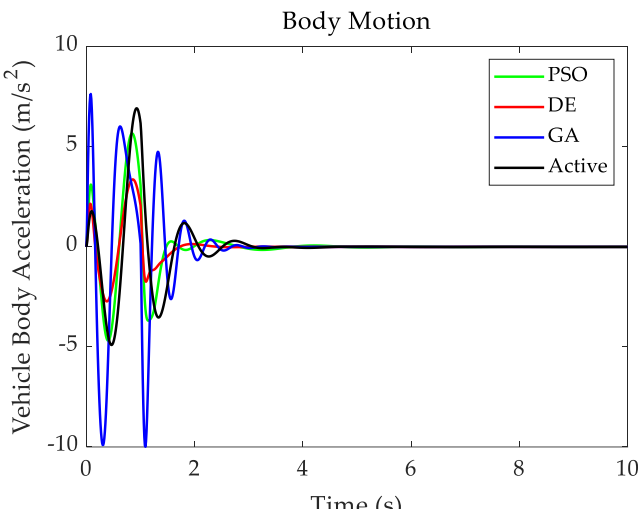


(c)

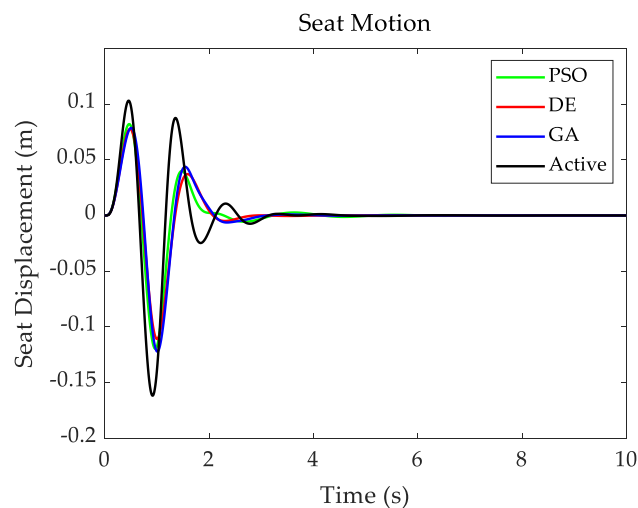
Fig. 10. The vehicle acceleration on a random road system (a) Body acceleration, (b) Seat acceleration, (c) Head acceleration of human body.



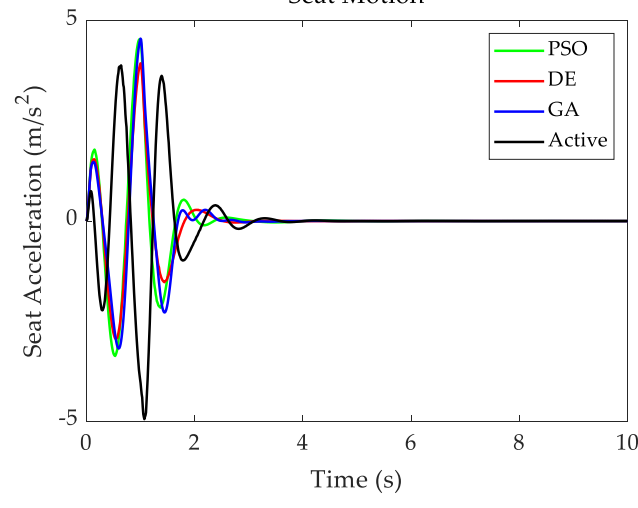
(a)



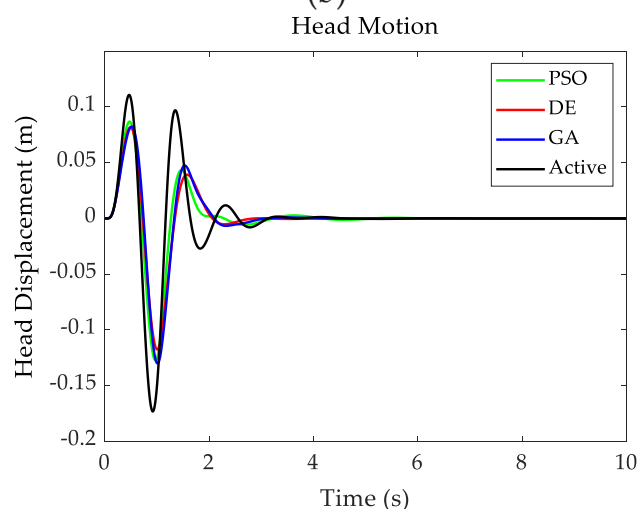
(a)



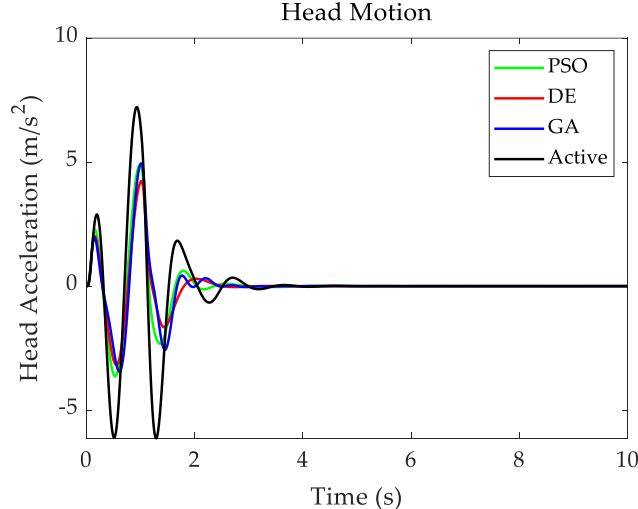
(b)



(b)



(c)



(c)

Fig. 11. Bump road simulation after optimizations (a) body vertical displacement, (b) seat vertical displacement, (c) head displacement of human body.

Fig. 12. Bump road simulation after optimizations (a) body acceleration, (b) seat acceleration, (c) head acceleration of human body.

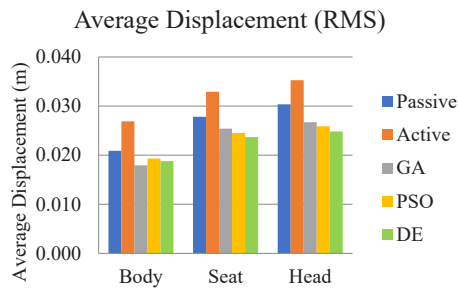


Fig. 13. Average displacement in harmonic motion after hitting a bump.

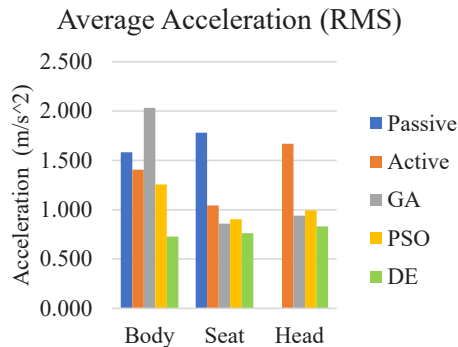


Fig. 14. Average acceleration of the system after hitting a bump.

the results obtained from GA optimization are more successful in vertical axis harmonic movements.

3.2. Optimization comparison

Simulations were performed using suspension values obtained with Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Differential Evolution (DE). The optimizations demonstrate the performance of each method. This comparison will determine which optimization method is more suitable for the vehicle model and provides the most efficient results. The main points considered during the comparisons are the acceleration rates of the vehicle body, seat, and head. RMS was used when defining the objective function in the optimizations, based on Eq. (51).

3.2.1. Bump model road simulations

The simulations for the bumpy road model use the bump road profile to compare the results of Active System, Genetic Algorithm (GA), Particle Swarm Optimization (PSO) and Differential Evolution (DE) optimization. The comparison focuses on changes in body vertical, seat and human head as an objective function, seat dynamics, and the human model's response.

As shown in the graphs in Fig. 11 and Fig. 12, the displacement and acceleration results obtained from Active System, Genetic Algorithm (GA), Particle Swarm Optimization (PSO) and differential evolution (DE) optimization.

Figs. 13 and 14 represent the average displacement and average acceleration of the system.

3.2.2. Random model road simulations

To compare optimizations for random road types, the changes in vehicle body vertical and biodynamic human model. Because of the random and uncertain nature of road movements in the system, improvement calculations were performed using RMS (Root Mean Square) values.

In Figs. 15 and 16, the first stage of the comparison of optimizations

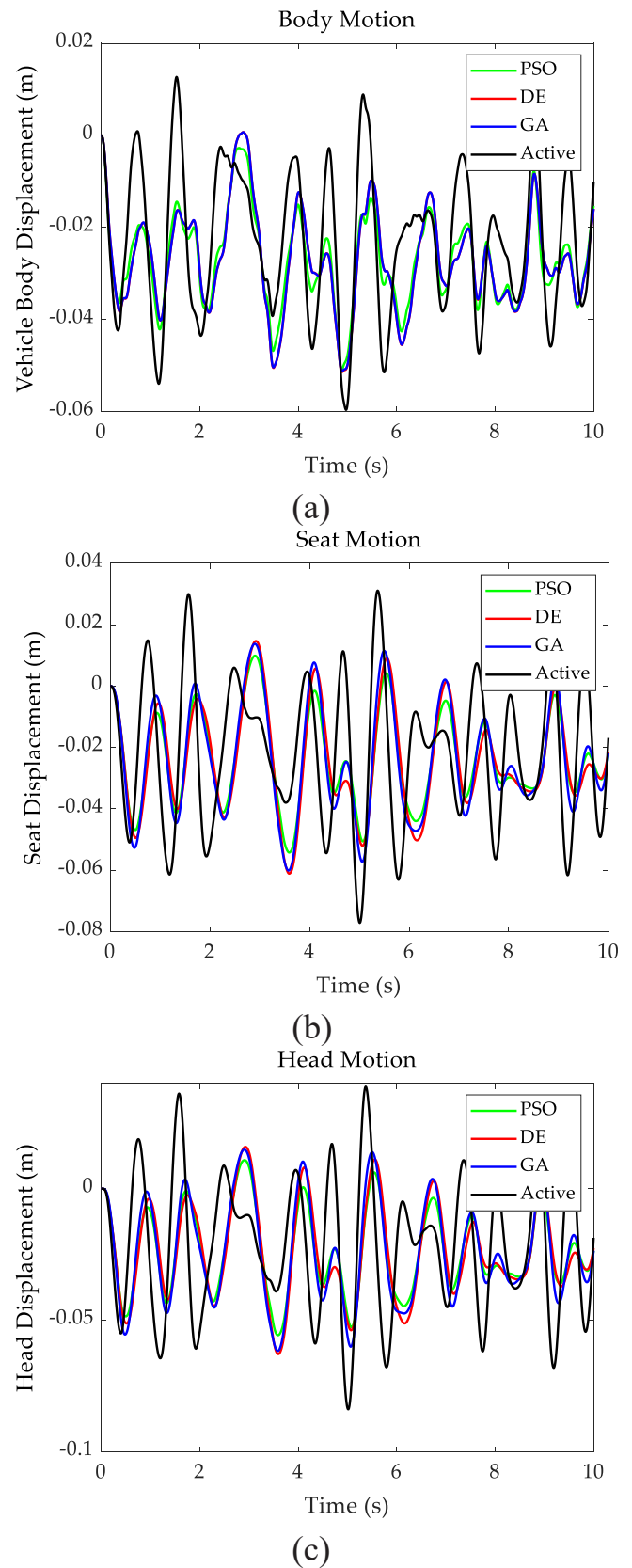


Fig. 15. Random road simulation after optimizations (a) Body vertical displacement, (b) Seat vertical displacement, (c) Head displacement of human body.

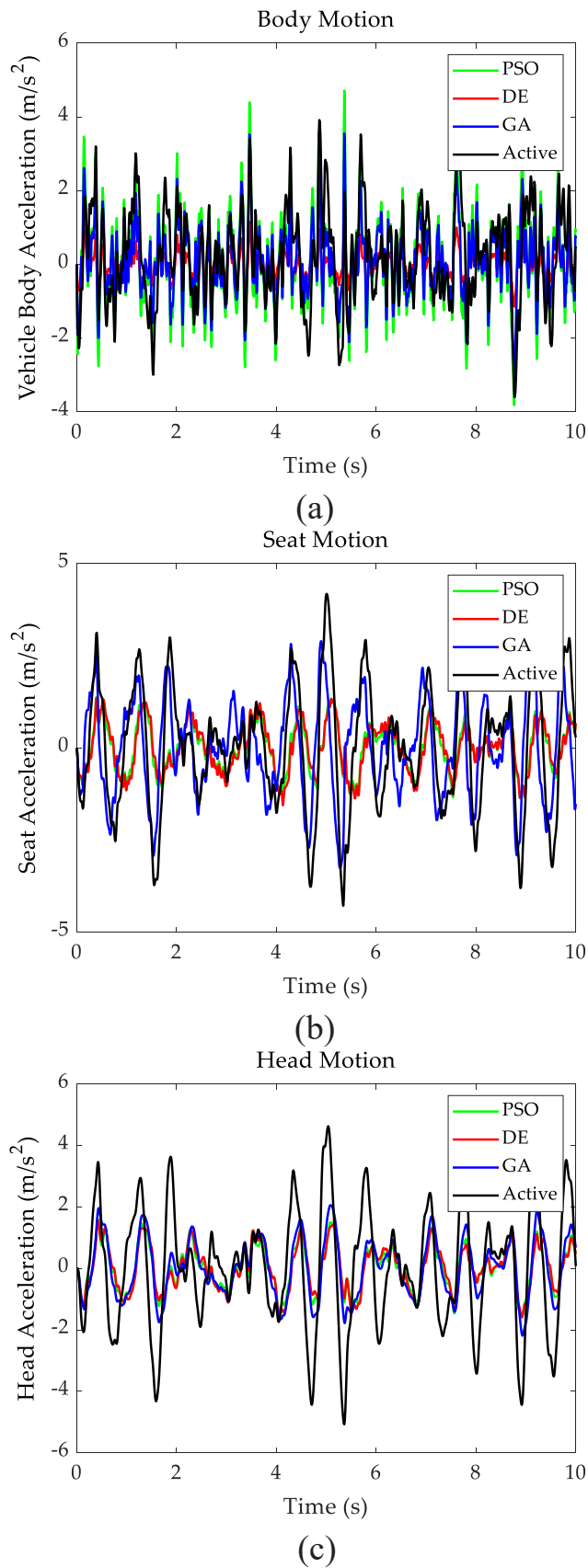


Fig. 16. Random road simulation after optimizations (a) body vertical acceleration, (b) seat vertical acceleration, (c) head acceleration of human body.

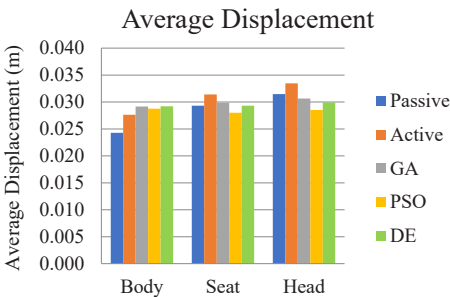


Fig. 17. Average displacement body, seat and human head on random road model.

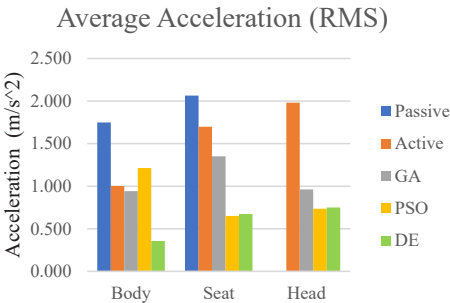


Fig. 18. Average accelerations body, seat and human head on random road model.

Table 8
Average changes in positions and their efficiency bump model.

	Body	Seat	Head
Passive	0.021	0.028	0.030
Active	0.027	0.033	0.035
DE	0.019	0.024	0.025
GA	0.018	0.025	0.027
PSO	0.019	0.025	0.026
η_{Active}	−29 %	−18 %	−16 %
η_{DE}	10 %	15 %	18 %
η_{GA}	14 %	9 %	12 %
η_{PSO}	8 %	12 %	15 %

for the random road model is based on the vertical displacement movements of the body, seat, and the head of the human model. The most critical aspect in Figs. 17 and 18 for the random road model optimizations is how much the human model is exposed to the movements.

In Figs. 17 and 18, the average movement average system displacement and accelerations are shown, calculated using RMS.

4. Discussion

In this section, the results of optimizations performed before and after optimization for the bump road model and random road model will be compared, and their impact on human health will be analyzed.

4.1. Bump model

When examining the displacement and acceleration movements in the bump-type road model, Figs. 7 and 8 show that the harmonic movements of the active system are better compared to the passive system. This can be seen in the values presented in Table 8 and Table 9.

According to the optimization results are examined, it is clearly observed in Tables 8 and 9 that DE optimization provides the most efficient results on acceleration but according to displacement PSO and

Table 9

Average changes in acceleration and their efficiency bump model.

	Body	Seat	Head
Passive	1.31370838	1.582	1.780
Active	1.405	1.043	1.668
DE	0.7281	0.7620	0.8305
GA	2.0315	0.8592	0.9407
PSO	1.2565	0.9036	0.9931
η_{Active}	21.1 %	34.1 %	6.3 %
η_{DE}	59 %	52 %	53 %
η_{GA}	–14.1 %	45.7 %	47.2 %
η_{PSO}	29.4 %	42.9 %	44.2 %

Table 10

Average accelerations of vehicle and human model and their efficiency random road model.

	Body	Seat	Head
Passive	1.43193985	1.749	2.065
Active	1.287	1.699	1.982
DE	0.3566	0.6737	0.7504
GA	0.9422	1.3507	0.9625
PSO	1.2137	0.6508	0.7355
η_{Active}	37.7 %	2.9 %	4.0 %
η_{DE}	83 %	61 %	64 %
η_{GA}	54.4 %	22.8 %	53.4 %
η_{PSO}	41.2 %	62.8 %	64.4 %

Table 11

Average Displacement of Vehicle and Human Model and Their Efficiency Random Road model.

	Body	Seat	Head
Passive	0.024	0.029	0.031
Active	0.028	0.031	0.033
DE	0.029	0.029	0.030
GA	0.029	0.030	0.031
PSO	0.029	0.028	0.029
η_{Active}	–14 %	–7 %	–6 %
η_{DE}	–20 %	0 %	5 %
η_{GA}	–20 %	–2 %	3 %
η_{PSO}	–18 %	4 %	9 %

GA optimization provides the best result. According to result objective functions DE provided best result.

4.2. Random model

In the comparison of the random-type road model, when comparing the data presented in Tables 10 and 12, improvements are observed compared to passive systems. In the results section, when comparing passive and active systems, improvements are noted in the passive system with the random road model for human health.

When comparing results in Table 10 accelerations according to optimizations result, PSO optimization provided the best results. This is followed by DE and the GA optimization result.

Table 11 presents the average displacement. When examining these results, PSO optimization provided the best outcomes, followed by DE optimization, GA optimization, for human health.

As a result, PSO optimization provided the best results for the random road type.

4.3. Driver health analysis

Ensuring driver health and comfort is crucial in vehicle design. This section compares the impact of different suspension systems on driver health, focusing on the effectiveness of GA, DE and PSO optimizations in reducing harmful vibrations and movements.

Table 12

Average Movement Head Accelerations of Human Model and Their Efficiency.

	RMS_{head}	NHR	PHR	LHR
Passive	2.065	0.5	0.5–1.7	1.7–24
Active	1.982	0.25	0.25–1	1–24
DE	0.7504	2.5	2.5–7.5	7.5–24
GA	0.9625	2	2–7	7–24
PSO	0.7355	3	3–8	8–24

Table 12 was prepared based on 2631–1:1997 standard, and the values in the table show the effects on the driver., the order of performance is PSO optimization, DE optimization, GA optimization active system, and passive system, respectively. Throughout the duration of the journey, the healthiest system is achieved with PSO optimization results. When comparing driving durations, the PSO optimization results show the longest No Health Risk (NHR) duration, with no health risks observed for 8 h. In the potential health risk range, the PSO optimization system is the last to enter and the last to exit this zone. Comparing the results, it is determined that in vehicles with passive systems, the health risk for the driver increases after 1.7 h.

5. Conclusions

The results and comparisons were conducted for two different road models. The outcomes of this simulation and optimization have been compared both numerically and visually. In the mathematical simulation using the bump road model, the analysis of graphs and tables indicates that DE optimization provides the most significant improvement for the human model, followed by PSO and GA optimizations. The best results for vehicle body acceleration reductions were achieved with DE optimization. DE optimization excelled in the bump model simulation due to its speed and ability to yield the best results. These findings illustrate the effectiveness of different optimization techniques in improving driving comfort and performance. The best result, based on the objective function, was obtained with PSO optimization, followed by DE and GA optimizations, respectively. However, active suspension systems did not demonstrate efficiency in the random road model. Considering all results, the ranking of effectiveness for human health on the random road model system is as follows: PSO optimization systems, DE optimization systems, and GA optimization systems, with active suspension systems not showing significant effectiveness. According to the bump road model, PSO optimization proves to be more effective than other systems in the random road model.

The obtained results are compared with similar studies where PD controller used for active suspension system. The maximum amplitudes of the accelerations were decreased up to %53 and similar achievements were also obtained by [7]. Comparing the results of [10], DE optimization has demonstrated the highest percentage improvement, confirming its effectiveness. However, for optimizations using a random road model, PSO has provided the best results, showcasing its adaptability to varying road conditions. The study also considers the findings of [14], which utilized a human model and applied fuzzy logic optimization via a PD-controlled system. The mathematical validation of these optimization techniques highlights their contribution to driver health and comfort under different conditions. It is evident that optimization systems effectively enhance suspension performance based on desired objective functions similar to [14]. Multi-objective optimization methodologies adopted in this study prove beneficial for active suspension systems, particularly in long-distance travel. The integration of real-time adaptive control mechanisms and different type of road validation is crucial for ensuring robustness. Expanding the optimization scope to include human health factors or specific objective functions can lead to the development of more efficient and adaptive suspension systems capable of addressing diverse operational challenges.

CRediT authorship contribution statement

Berk Aydoğan: Writing – review & editing, Writing – original draft, Validation, Project administration, Formal analysis, Resources, Methodology, Data curation, Visualization, Investigation. **Ahmet Yildiz:** Visualization, Supervision, Conceptualization.

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